OFFICIAL CLINICAL DOSSIER · PRECISION CAPILLARY MEDICINE

TrichoTest™-Guided Hair Treatments Collection

Integrated Multimodal Ecosystem: TrichoSol™ Compounded Formulations, Follicular Peptides & Nutriceuticals

DNA: 48 SNPs / 13 Genes | TrichoSol™ Vehicle | GHK-Cu 50 mg | Baicapil™ 2% & Collagen | HTMA ICP-MS/MS

1. Genetic Foundation and Clinical Rationale of TrichoTest™

The TrichoTest™ pharmacogenomic assay analyzes 48 single nucleotide polymorphisms (SNPs) across 13 genes governing hair follicle metabolism via DNA microarray and qPCR (PTGFR/PTGIR prostaglandin receptors, 5-alpha-reductase SRD5A1/SRD5A2, follicular sulfotransferase SULT1A1, androgen receptor AR, nitric oxide synthase NOS3, and perifollicular inflammation CRTH2/PTGDR2).

Standard single-drug therapies often fail because patients with low SULT1A1 activity cannot sulfoconjugate Minoxidil, while those with PTGFR mutations need targeted prostaglandin modulation. This collection converts individual genetic profiles into personalized compounding and synergistic follicular care.

2. Multimodal 5-Pillar Treatment Architecture

Pillar	Clinical Component	Mechanism of Action & Pharmacodynamic Rationale
Pillar 1	**Compounded Formulations in TrichoSol™ / TrichoFoam™**	Bioactivated Minoxidil, Finasteride/Dutasteride, Spironolactone, Prostaquinon™ and biomimetic peptides (IGrantine-F1™) formulated in a patented 0% alcohol, 0% propylene glycol base, ensuring superior epicutaneous tolerability.
Pillar 2	**Follicular Peptide Bioregulation (GHK-Cu / TB-500)**	GHK-Cu (copper tripeptide-1 50 mg) stimulates dermal papilla fibroblasts, downregulates catagen-driver TGF-Beta-1, and promotes VEGF/bFGF microvascular angiogenesis without hormonal competition.
Pillar 3	**Extracellular Matrix & Follicle Care (Baicapil™ & Collagen)**	Shampoo & Conditioner formulated with Baicapil™ 2% (-60.6% hair loss arrest, +49% density in 84-day RCT) and native fish tropocollagen protecting against cuticle lipid depletion and infundibular detachment.
Pillar 4	**Toxicological & Mineral HTMA Diagnostics**	ICP-MS/MS hair tissue mineral spectrometry measuring 33 intracellular elements and heavy metals (Pb, Hg, Cd, As). Evaluates 90-day metabolic ratios (Ca/Mg, Na/K, Zn/Cu) that can sabotage topical drug response.
Pillar 5	**Systemic Matrix Fortification & Analytical Monitoring**	Lyophilized native tropocollagen oral capsules with marine algae for systemic ECM remodeling, aligned with androgenic follicular panel telemetry (Free DHT, Ferritin >70 ng/mL, Zinc, Vitamin D3).

Technological Superiority of Patented TrichoSol™ Vehicle

Traditional alcohol and propylene glycol solutions trigger scalp erythema, contact dermatitis, and lipid barrier damage leading to high discontinuation rates. TrichoSol™ features the patented TrichoTech™ phytocomplex (100% alcohol and glycol free), tripling follicular drug absorption while protecting epicutaneous scalp balance.

PILLAR 1 · OFFICIAL COMPOUNDED FORMULATIONS

Precision Compounded Formulations in the Collection

Custom formulations prepared under individual medical prescription, aligned with patient genomics.

Formulation & Vehicle	Active APIs & Strength	Genetic & Clinical Target	Administration Protocol
TrichoSol™ Androgenic Duo (100 mL Solution)	• **Minoxidil 4%** • **Spironolactone 1%** • **L-Arginine 1.5%**	Heightened DHT sensitivity + impaired microvascular perfusion (NOS3). Follicular receptor antagonism.	Apply 1 mL at night onto dry scalp with gentle massage. Leave overnight; wash next morning.
TrichoFoam™ Triple Reactivator (100 mL Foam)	• **Minoxidil 4%** • **Dutasteride 0.25%** • **IGrantine-F1™ 0.5%**	5-alpha-reductase type I/II hyperactivity (SRD5A1/2) with severe hair bulb miniaturization.	Apply 2 pumps of foam at night; massage into thinning crown and hairline.
TrichoSol™ Anti-PGD2 Shield (100 mL Solution)	• **Prostaquinon™ 3%** • **Spironolactone 1%** • **IGrantine-F1™ 0.25%**	High PGD2 pathway activation and CRTH2 receptor sensitivity. Suppresses micro-inflammatory arrest.	Apply 1 mL daily before bedtime. Allow full absorption.
TrichoSol™ Female Balance (100 mL Solution)	• **17-Alpha-Estradiol (Alfatradiol) 0.05%** • **IGrantine-F1™ 0.5%**	Female androgenetic pattern or intolerance to systemic anti-androgens. Zero systemic feminization.	Apply 1 mL once daily with calibrated dropper to mid-scalp and diffuse thinning areas.
TrichoOil™ Pre-Wash Nutrient (30 mL Scalp Oil)	• **Panax Ginseng 2%** • **Ginkgo Biloba 2.5%** • **Vitamin E (Tocopherol) 5%**	Oxidative scalp stress, cutaneous lipid loss, or damaged epidermal barrier from chronic topicals.	1–2 times weekly: massage 3–5 min, leave for 10 min before shampooing.

Genetic Polymorphism and API Target Correlation

Genetic Pathway	Genes / SNPs	Recommended API	Pharmacodynamic Rationale
Prostaglandin Axis	**PTGFR / PTGIR**	**Latanoprost 0.005% / Prostaquinon™**	Prolongs anagen phase and recruits dermal papilla vessels.
Androgen Metabolism	**SRD5A1 / SRD5A2**	**Dutasteride / Finasteride / 17-Alpha-Estradiol**	Competitive 5-alpha-reductase inhibition without systemic side effects.
Sulfotransferase	**SULT1A1**	**Minoxidil in TrichoSol™ or Oral**	Identifies whether patient responds to topical vs oral Minoxidil.
Nitric Oxide & Perfusion	**NOS3 / eNOS**	**L-Arginine 1.5% + GHK-Cu**	Peribulbar vasodilation, microvascular O2 and nutrient feed.
Follicular Inflammation	**CRTH2 / PTGDR2**	**Cetirizine / Prostaquinon™**	Inhibits PGD2-mediated follicular arrest and miniaturization.

PILLARS 2, 3, 4 & 5 · BIOREGULATION, CARE & ROUTINE

Peptides, Scalp Protection & Patient Daily Routine

Pillar 2: Copper Peptide Bioregulation (GHK-Cu 50 mg)

GHK-Cu (Copper Tripeptide-1, 50 mg sterile lyophilized vial) acts as a powerful dermal papilla bioregulator. It downregulates Transforming Growth Factor Beta-1 (TGF-Beta-1), the primary cytokine driving catagen transition and miniaturization. Additionally, it stimulates microvascular angiogenesis via VEGF and bFGF upregulation without hormonal receptor competition.

Pillar 3: Follicular Barrier Protection & Native Tropocollagen

Chronic use of topical Minoxidil and antiandrogens frequently leads to scalp desquamation and cuticle lipid depletion. The complementary trichological protocol delivers:
• Strengthening Shampoo (200 mL): Featuring Baicapil™ 2% (Scutellaria baicalensis, soy & wheat sprouts), micronized diosmin, and native fish tropocollagen. RCT clinical trials confirm -60.6% hair loss arrest and +49% hair density increase over 84 days (anagen/telogen ratio rises from 2.3 to 4.7).
• Strengthening Conditioner (200 mL): Seals cuticle scales and reinforces infundibular follicular anchorage.
• Native Tropocollagen Oral Capsules: Lyophilized native tropocollagen with Fucus vesiculosus for systemic ECM rebuilding.

Pillar 4: Intracellular Mineral & Heavy Metal Toxicology (HTMA)

Hair tissue mineral analysis via ICP-MS/MS mass spectrometry quantifies 33 essential minerals and toxic heavy metals (Pb, Hg, Cd, As). Unlike transient serum levels, it evaluates 90-day intracellular storage and metabolic ratios (Ca/Mg, Na/K, Zn/Cu) that can otherwise sabotage topical response.

Patient Application Guide (Weekly Protocol Schedule)

Timing	Clinical Step	Product / Formulation	Key Instructions
Morning (3–4x weekly)	**Follicular Hygiene & Stem Cell Stimulation**	**Strengthening Shampoo with Baicapil™ 2%**	Massage onto wet scalp for 2–3 minutes to cleanse follicular ostia. Rinse with lukewarm water.
Morning (Post shampoo)	**Cuticle Sealing & Hair Shaft Protection**	**Strengthening Cuticle Conditioner**	Apply to mid-lengths and ends. Leave for 2 minutes and rinse thoroughly.
Breakfast (Daily)	**Systemic Matrix Fortification**	**Native Tropocollagen (2 capsules daily)**	Take 2 capsules with water at mealtime to supply bioactive tropocollagen oligopeptides and marine minerals.
Night (Daily)	**Precision Follicular Compounding**	**Custom Formula in TrichoSol™ / TrichoFoam™**	Apply 1 mL (or 2 pumps) onto clean, dry scalp with gentle finger massage. Leave on overnight; do not rinse immediately.
Weekly (1–2x weekly)	**Pre-Wash Lipid Barrier Conditioning**	**TrichoOil™ Scalp Care Oil (30 mL)**	Massage onto scalp 10 minutes prior to washing to prevent lipid barrier dryness and vehicle irritation.

Medical Notice & Compounding Quality Guarantee

Compounded medications in this collection are prepared under individualized medical prescription in certified cleanroom compounding facilities compliant with Good Compounding Practices and European Pharmacopoeia standards. Topical adjuvant formulations comply with EU Cosmetic Regulation 1223/2009 (CPNP registered). HTMA intracellular trace mineral analyses are performed using ICP-MS/MS spectrometry in ISO/IEC 17025 accredited laboratories.

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